



Computer Vision Lab

Part 1: Setup and Data Exploration

- Dataset: Fashion MNIST - 70,000 grayscale images of fashion items, 28×28 pixels
- 10 classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot
- Split: 60,000 training images, 10,000 test images
- Load via `keras.datasets.fashion_mnist.load_data()`
- Each image: single channel (28, 28) with pixel values in [0, 255]
- After division by 255: values normalized to [0.0, 1.0]
- After zero-centering (mean=0.1307, std=0.3081): normalized distribution
- Add channel dimension for TensorFlow: (28, 28, 1)

Part 2: Data Augmentation

- Use `keras.preprocessing.image.ImageDataGenerator` for augmentation
- Training augmentation: `rotation_range=30` degrees ($\pm 30^\circ$)
- Validation/test: no augmentation (only normalization)
- Implement `rotate_by_multiple_90()` to rotate images by 0° , 90° , 180° , 270° (using `np.rot90`)
- Visualize augmented samples: 4×4 grid showing same image with different random rotations
- Compare pixel statistics: rotation preserves mean and std (only geometric transformation)

Part 3: CNN Architecture

- Use `keras.Sequential` API for simplicity
- 3 convolutional blocks (not a class-based model)
- Block structure: `Conv2D(filters, kernel_size=3, padding='same')` → `BatchNormalization` → `ReLU` → `MaxPooling2D(2)`
- Filter progression: 32 → 64 → 128 filters
- Spatial dimensions: (28, 28) → (14, 14) → (7, 7)
- Classifier head: `Flatten` → `Dense(256)` → `ReLU` → `Dropout(0.4)` → `Dense(10)` [logits]
- Print `model.summary()` and count total trainable parameters

Part 4: Training and Evaluation

- Optimizer: Adam (learning_rate=1e-3)
- Loss: SparseCategoricalCrossentropy (from_logits=True)
- Train using model.fit() with both augmented and non-augmented datasets
- Create TensorFlow datasets using tf.data.Dataset.from_tensor_slices() with shuffle and batch
- Plot: training loss and train/val accuracy across 20 epochs
- Evaluate on rotated test images at angles: 0°, 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330°
- Compare: plot accuracy-vs-angle curves for augmented vs. non-augmented models

Lab Assignment

Complete the following tasks using CIFAR-10 dataset and a simpler CNN architecture.

Task 1: Load and Explore CIFAR-10

1. Load CIFAR-10 using `keras.datasets.cifar10.load_data()`
2. Print shape of training and test data
3. Display a 3×3 grid of random images with their class names

Task 2: Basic Preprocessing

4. Normalize pixel values by dividing by 255.0
5. Compute and print mean and std of the training set
6. Create train/val split (use 50,000 training images)

Task 3: Build a Simple CNN

7. Build a Sequential model with 2 convolutional blocks
8. Block structure: `Conv2D(filters, 3) → ReLU → MaxPooling2D(2)`
9. Use 32 filters in block 1, 64 filters in block 2
10. Add `Flatten → Dense(128) → ReLU → Dense(10)` classifier
11. Print `model.summary()`

Task 4: Train and Evaluate

12. Compile model with Adam optimizer (`lr=0.001`) and categorical crossentropy loss
13. Train for 10 epochs with `batch_size=64`
14. Plot training and validation accuracy on the same chart
15. Evaluate on test set and print accuracy